# Lab 6 — Fit and Compress

**Friday, December 4 (~20 min)**

Last updated: October 29, 2026.

Friday, December 4 is **project launch** (capstone) with Lecture 33 (SVD) and Quiz 11. This lab is the same-day LS + truncated-SVD software warm-up — fine to meet SVD in lecture and practice it here. Quiz 11 is Week 13 eigenvalues — **not** this lab.

**Credit:** Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

## Learning objectives

1. Solve a small least-squares / fit problem in software.
2. Run a truncated SVD on a small matrix and rebuild a rank-$k$ approximation.
3. Compute residual size and storage count $k(m+n+1)$ for two values of $k$.

## Math connections

Least squares (Lectures 38–39); SVD and the orthonormal / LS bridge (Lecture 33); warm-up for the image-compression capstone.

## Key syntax (this lab)

| Task | Octave | Python |
|---|---|---|
| Least squares | `A\b` (overdetermined) | `np.linalg.lstsq(A,b,rcond=None)` |
| Full SVD | `[U,S,V]=svd(A)` | `U,s,Vt=np.linalg.svd(A,full_matrices=False)` |
| Rank-$k$ rebuild | `U(:,1:k)*S(1:k,1:k)*V(:,1:k)'` | `U[:,:k] @ np.diag(s[:k]) @ Vt[:k,:]` |
| Frobenius norm | `norm(A-Ak,'fro')` | `np.linalg.norm(A-Ak,'fro')` |

## Ops strip